THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE TWO-CHAPTER TEST SET B

Chapters 5 & 6: **Describing Motion Around Us, How Forces Affect Motion** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Displacement is: 1

(a) always positive
(b) the total path length covered
(c) the shortest straight-line change in position with a direction
(d) always equal to distance

2. The slope of a position–time graph gives: 1

(a) distance only **(b)** displacement **(c)** acceleration **(d)** velocity

3. A horizontal line on a velocity–time graph shows: 1

(a) constant velocity, zero acceleration **(b)** rest
(c) constant acceleration **(d)** decreasing velocity

4. A body thrown vertically upward and falling back to the thrower's hand has displacement: 1

(a) the maximum height **(b)** equal to the distance
(c) twice the height **(d)** zero

5. Which graph shows a body moving with uniform acceleration? 1

(a) a horizontal line on a velocity–time graph
(b) a straight inclined line on a velocity–time graph
(c) a vertical line on a position–time graph
(d) a horizontal line on a position–time graph

6. A passenger standing in a bus falls backwards when the bus suddenly starts because of: 1

(a) friction on the bus **(b)** weight
(c) inertia of rest **(d)** inertia of motion

7. Action and reaction forces do not cancel because they: 1

(a) act at different times **(b)** act in the same direction
(c) are unequal **(d)** act on different objects

8. Grooves on shoe soles and tyre treads are made to: 1

(a) reduce friction **(b)** reduce weight
(c) increase speed **(d)** increase friction

9. For the same force, a body with larger mass has: 1

(a) zero acceleration **(b)** larger acceleration
(c) smaller acceleration **(d)** the same acceleration

10. A body of mass 2 kg is acted on by two forces, 8 N east and 3 N west. Its acceleration is: 1

(a) 2.5 m s^{-2} west **(b)** 5.5 m s^{-2} east **(c)** 1.5 m s^{-2} east **(d)** 2.5 m s^{-2} east

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 10–15 words each.

11. Write the three kinematic equations for uniformly accelerated motion. 2

12. A car starting from rest has acceleration 3 m s^{-2}. Find its velocity after 4 s. 2

13. Find the net force when 10 N and 6 N act on a block in opposite directions. 2

14. Why does a fielder draw his hands back while catching a fast ball? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 30–40 words each.

15. Why is it important to keep a safe distance from the vehicle ahead? Relate to stopping distance. 3

16. How can a spacecraft in space with negligible gravity slow down? 3

17. Explain uniform circular motion with an example. Is the motion accelerated? 3

OR

A body moves round a circle of radius 14 m in 22 s. Find the distance, displacement after one round and speed (π = 22/7). 3

18. Two blocks of masses 2 kg and 3 kg joined by a string are pulled on a smooth floor by a 10 N force. Find the acceleration and the tension in the string. 3

OR

Differentiate between mass and weight. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. A child on a merry-go-round of radius 7 m completes 5 revolutions in 110 s (π = 22/7). 5

(i) Find the distance.
(ii) Find the displacement.
(iii) Find the average speed.
(iv) Find the average velocity.

OR

Case study: A ball is dropped from a height; its velocity at successive seconds is 9.8, 19.6, 29.4 m s^{-1}. 5

(i) Find the acceleration in each second.
(ii) Is it constant?
(iii) Name this acceleration.
(iv) Find the distance fallen in 3 s.

20. (i) Explain the system approach for two boxes of m_1 and m_2 pulled by force F. 5
(ii) Find the acceleration for m_1 = 2 kg, m_2 = 3 kg, F = 10 N.
(iii) Find the tension.

OR

(i) Distinguish balanced and unbalanced forces. 5
(ii) Two teams pull a rope with 400 N and 300 N. Find the net force and direction.
(iii) Which team wins and why?

END OF QUESTION PAPER — Total: 40 Marks